CIGPO: Contextual Information-Gain Policy Optimization
for Multi-Turn Evidence-Reading LLM Agents

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Abstract

Training multi-turn evidence-reading agents with outcome-only reinforcement learning is unstable because intermediate turns receive little direct credit. In HotpotQA experiments with Qwen2.5-3B-Instruct, GRPO initially improves (standard F1 0.430) but subsequently collapses to 100% format-violating outputs. Training-log diagnosis reveals a zero-advantage lock-in mechanism: all sampled trajectories receive the minimum format penalty (-2.0), group-relative advantages vanish, and the policy-gradient loss becomes zero—an optimization deadlock. We propose a variance-injection strategy: by assigning per-turn rewards to intermediate evidence-reading turns, we prevent the group reward distribution from collapsing to a single value—preserving the variation that GRPO's group-relative advantage requires. **C**ontextual **I**nformation-**G**ain **P**olicy **O**ptimization (CIGPO) implements this strategy using the marginal increase in the frozen reference model's log-likelihood of the ground-truth answer as the per-turn signal. With separate normalization of IG and F1 rewards and an IG-weight curriculum, CIGPO reaches a standard F1 of 0.518 on HotpotQA at the 3B scale (from 0.252 base; +105%), compared with 0.430 for the best GRPO checkpoint and 0.000 for the final GRPO checkpoint. CIGPO maintains meaningful reward variance and avoids zero-advantage lock-in throughout training. As a focused empirical study, these results suggest that injecting reward variance at intermediate turns can stabilize GRPO-based multi-turn evidence-reading RL in this setting.

1 Introduction

Large language models (LLMs) are increasingly deployed as autonomous agents that interact with external tools and knowledge sources over multiple turns [1–3]. A canonical task in this paradigm is *multi-turn evidence reading*: given a question, the agent decides which documents to read, retrieves evidence, and synthesizes an answer across multiple rounds of retrieval and reasoning.

Reinforcement learning (RL) is a natural framework for training such agents. Recent advances in RL for LLMs, particularly Group Relative Policy Optimization (GRPO) [4], have shown impressive results in math reasoning and code generation by optimizing a sparse outcome reward (e.g., final answer correctness). However, we observe a critical stability issue when applying GRPO to multi-turn evidence reading: GRPO initially improves (standard F1 reaches 0.430 at step 50 on Qwen2.5-3B-Instruct) but subsequently collapses, with the model degenerating to 100% format-violating outputs by step 150. Training logs at the point of collapse show that critic score mean converges to exactly -2.000, the zero-advantage group ratio reaches 1.0, and the policy-gradient loss becomes exactly 0.000. The resulting entropy loss spikes to $\sim$4809, consistent with garbled high-entropy output distributions from which the model cannot recover without an external learning signal.

We identify the root cause as an **optimization deadlock** arising from the credit assignment problem in multi-turn RL: when only the final turn receives a non-zero reward, intermediate turns lack a learning signal. The policy drifts toward degenerate behaviors—such as omitting required XML tags or generating garbled output—that, while scoring the minimum format penalty (-2.0), produce zero advantage when all group members also produce violations. Without variation in group rewards, GRPO provides no meaningful gradient, and the policy collapses into a deadlock from which it cannot recover.

To address this, we adopt a **variance-injection** principle: the key requirement is to ensure that the reward distribution across GRPO group members retains non-zero variance, so that group-relative advantages never vanish. As one concrete implementation, we propose **Contextual Information-Gain Policy Optimization** (CIGPO). CIGPO is a lightweight extension of GRPO that injects variance at intermediate turns via *reference-based contextual information gain*. Information gain quantifies how much each evidence-reading turn improves the reference model's confidence in the correct answer:

$$r_t^{\mathrm{IG}} = \log p_{\mathrm{ref}}(y^* \mid q, e_{\leq t}) - \log p_{\mathrm{ref}}(y^* \mid q, e_{<t}) \tag{1}$$

where p_{ref} is the frozen reference model (the same model used for KL regularization in GRPO), y^* is the ground-truth answer, q is the question, and $e_{\leq t}$ denotes evidence read up to turn t.

CIGPO adapts information-gain-based credit assignment [14] to a controlled evidence-reading environment. Unlike search-based settings where the model formulates free-form queries, CIGPO operates in a closed evidence pool: at each turn the agent selects and reads pre-indexed evidence blocks via structured `Read[block_id]` actions. The IG reward is assigned as a sparse turn-level signal—placed on the last token of each intermediate turn—while the standard F1 reward is assigned to the last token of the final turn.

Our contributions are:

- We diagnose the training instability of outcome-only GRPO in multi-turn evidence reading as an optimization deadlock, showing that GRPO initially improves but then collapses to 100% format violations via zero-advantage lock-in in our Qwen2.5-3B HotpotQA setting.
- We propose a variance-injection approach to multi-turn credit assignment, and implement it as CIGPO—which adapts reference-based contextual information gain to evidence-reading turns—demonstrating that it stabilizes training at the 3B scale, improving standard F1 from 0.252 to 0.518 on HotpotQA.
- Through analysis of per-turn IG values and training-log metrics, we provide evidence that IG rewards are correlated with successful evidence use (mean cumulative IG of 4.06 for correct vs. 2.10 for wrong trajectories) and that CIGPO maintains meaningful reward variance throughout training, preventing the zero-advantage lock-in observed in GRPO.

This is a focused empirical study on a single dataset (HotpotQA) with a Qwen2.5-3B-Instruct model under constrained hardware (2×24GB GPUs). The goal is not a universal solution for all multi-turn RL settings, but to document a specific failure mode—zero-advantage variance collapse—and demonstrate that a turn-level reward signal can prevent it in this setting.

Table 1: Comparison of IGPO and CIGPO. CIGPO adapts IG-based credit assignment to controlled evidence-reading environments.

Aspect	IGPO	CIGPO
Action type	Search query generation	Evidence block selection
Action space	Open-ended web search	Closed evidence pool
Environment	Web / search engine	Local pre-indexed documents
Action format	Free-form search query	Read[evidence_id]
IG computation	IG over search context	IG over evidence context
Base optimizer	Various RL algorithms	GRPO
Domain	Multi-turn search agents	Multi-turn evidence-reading agents

2 Related Work

RL for LLM Agents. Reinforcement learning has become a standard approach for aligning and improving LLMs. Proximal Policy Optimization (PPO) [7] and its variants have been widely adopted for RLHF [8, 9]. Direct Preference Optimization (DPO) [6] offers a simpler alternative by optimizing preferences directly, though it does not natively handle multi-turn credit assignment. More recently, GRPO [4] eliminates the value network by using group-level relative advantages, as in DeepSeekMath and DeepSeek-R1 [10]. These methods have proven effective for math and code generation where a single-turn response suffices. Our work extends GRPO to multi-turn settings by adding per-turn credit assignment.

Multi-Turn Agent Training. Several works explore RL training for multi-turn agent interactions. ReAct [1] and Toolformer [2] enable LLMs to use tools in a multi-turn fashion but do not use RL for optimization. Agent Q [11] and StreamBench [12] apply RL to multi-turn agent tasks but rely on value-based methods or Monte Carlo tree search. [13] uses EM-based search to provide turn-level signals for search agents.

Information Gain as Reward. Information gain has a long history in feature selection and active learning [15]. In the context of LLM agents, IGPO [14] proposes using information gain to reward search queries in web-based information-seeking agents. Our work, CIGPO, adapts this idea to a different setting: controlled evidence reading from a closed document pool rather than open-ended web search. Table 1 summarizes the key differences.

Credit Assignment in RL. The credit assignment problem is fundamental in RL [18, 19]. In sequence-level RL for LLMs, prior work has explored token-level reward redistribution using learned critics [16] and process reward models [17]. CIGPO offers a simpler alternative: the IG signal provides immediate, per-turn feedback using only a frozen reference model, without requiring a separately trained reward model.

3 Preliminaries

3.1 Group Relative Policy Optimization (GRPO)

GRPO [4] optimizes a policy π_θ by comparing responses within a group. For each prompt x, a group of N responses $\{o_i\}_{i=1}^N$ is sampled from the current policy. Each response receives a reward r_i, and advantages are computed using group-level normalization:

$$A_i = \frac{r_i - \mu_{\text{group}}}{\sigma_{\text{group}}} \tag{2}$$

where μ_{group} and σ_{group} are the mean and standard deviation of rewards within the group. The policy is then updated via a clipped surrogate objective:

$$\mathcal{L}_{\text{GRPO}} = \mathbb{E}\left[\min\left(\rho_i A_i,\ \text{clip}(\rho_i, 1-\epsilon, 1+\epsilon)A_i\right) - \beta\, D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\right] \tag{3}$$

where $\rho_i = \pi_\theta(o_i|x)/\pi_{\text{old}}(o_i|x)$ is the importance weight, ϵ is the clipping threshold, and β controls the KL penalty toward a reference policy.

3.2 Multi-Turn Evidence Reading

We formalize multi-turn evidence reading as a sequential decision process. Given a question q and a set of candidate evidence documents $\mathcal{E} = \{e_1, \ldots, e_K\}$, the agent interacts for up to T turns. At each turn t, the agent:

1. Reasons about what information is needed (in a <think> block)
2. Calls a tool to read a document (via <tool_call> with a structured Read[block_id] action)
3. Receives evidence content (in a <tool_response> block)

After the final turn (or earlier if the agent chooses), the agent produces a final answer in an <answer> tag. The answer is scored against the ground truth using token-level F1.

In standard GRPO-based training, only the final turn's answer token receives a non-zero reward. All intermediate turns—the reasoning and tool-calling decisions—receive zero reward.

4 CIGPO Method

CIGPO extends GRPO with three components designed for multi-turn credit assignment. The unifying principle is variance injection: GRPO's group-relative advantage requires reward variation within each group to produce a meaningful gradient. When all trajectories in a group receive identical rewards—as when every sample produces a format violation—advantages vanish and the policy receives no learning signal. CIGPO prevents this by ensuring intermediate evidence-reading turns carry non-zero, varied rewards. Figure 1 contrasts the reward structure of GRPO and CIGPO.

4.1 Per-Turn Information Gain Reward

The core of CIGPO is rewarding each evidence-reading turn with its information gain. For turn t (where $1 \leq t < T$), we define:

$$r_t^{\text{IG}} = \log p_{\pi_{\text{ref}}}(y^* \mid q, e_1, \ldots, e_t) - \log p_{\pi_{\text{ref}}}(y^* \mid q, e_1, \ldots, e_{t-1}) \tag{4}$$

where $p_{\pi_{\text{ref}}}$ is the probability under the frozen reference model. We use the *reference* model (rather than the current policy) for three reasons: (1) it avoids non-stationary reward drift as the policy updates; (2) it is consistent with the reference model already used for KL regularization in GRPO; and (3) it makes IG computation stable and reproducible, as the reference model is frozen throughout training.

The final turn T receives the standard F1 reward:

$$r_T^{\text{F1}} = \text{F1}(\text{extract\_answer}(o_T), y^*) \tag{5}$$

Reward placement. IG rewards are assigned as sparse turn-level signals: the IG value is placed on

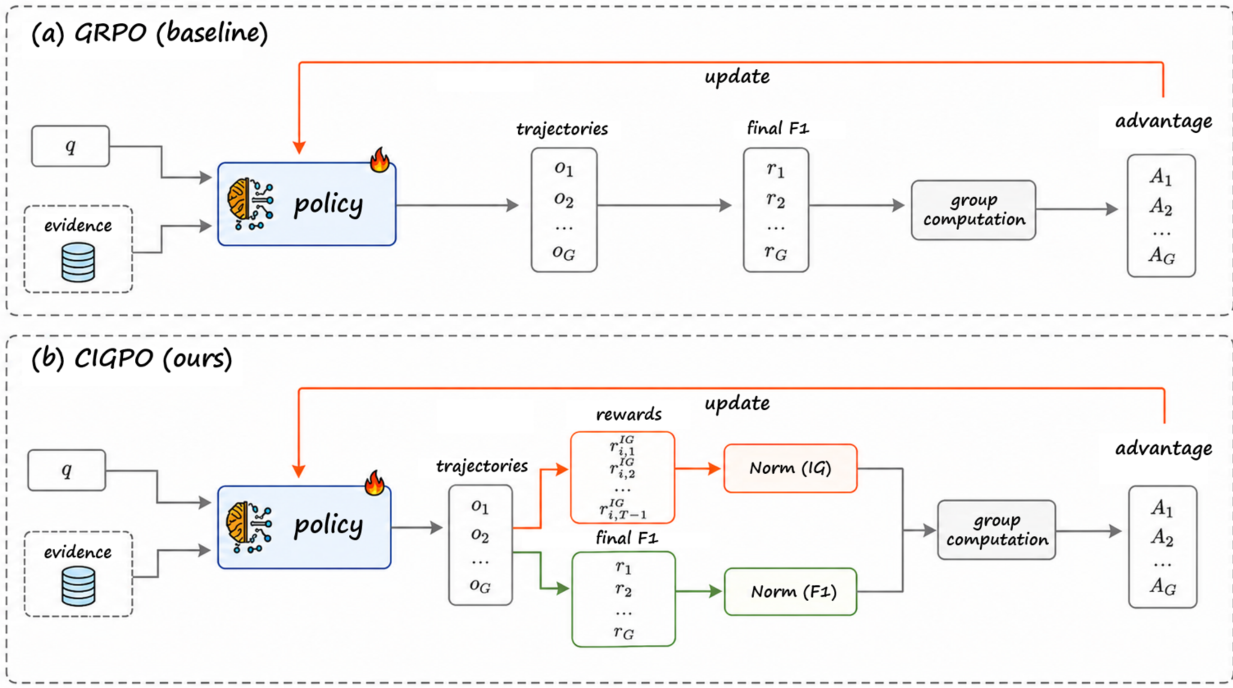

Figure 1: Overview of the CIGPO training framework. Given a question and a pool of pre-indexed evidence documents, the policy model (trained via GRPO with IG-augmented rewards) selects and reads evidence blocks across multiple turns. At each intermediate turn t, the contextual information gain $\Delta_t = \log p_{\text{ref}}(y^* \mid q, e_{\leq t}) - \log p_{\text{ref}}(y^* \mid q, e_{<t})$ is computed using the frozen reference model and assigned as a per-turn reward. The final turn receives the standard F1 reward against the ground-truth answer. Separate group normalization is applied to IG and F1 rewards before advantage computation.

the last token of each intermediate turn (terminal token reward placement), and the F1 reward is placed on the last token of the final turn. All other tokens receive zero reward. No turn-level discounted returns are used. This creates a sparse but turn-granular reward structure that provides feedback at each decision point without requiring dense per-token credit assignment.

4.2 Prealigned Vectorized GT LogProb

Computing IG naively requires T separate forward passes of the reference model (one per turn prefix). We accelerate this via *prealigned vectorized computation*: all turn prompts are padded to a uniform length and concatenated into a single batch. The reference model computes $\log p(y^* \mid \cdot)$ for all T turns in one batched forward pass, achieving a $\sim 3\times$ speedup over sequential computation.

4.3 Separate GRPO Group Normalization

A critical design choice is *how* IG and F1 rewards are normalized before computing advantages. GRPO normalizes rewards within a group using Equation 2. However, IG and F1 operate on fundamentally different scales: IG values are in the nat scale (mean ≈ 8, range $[-17, +29]$ in practice), while F1 is bounded in $[0, 1]$ (or receives a format penalty of -2.0).

We use **separate normalization**: IG and F1 rewards are normalized independently within each group. For a group of N trajectories, let $\mathcal{R}_{\text{IG}}$ be the set of all intermediate-turn IG rewards and $\mathcal{R}_{\text{F1}}$ be the set of final-turn rewards. Advantages are computed as:

$$A_t^{\mathrm{IG}} = \frac{r_t^{\mathrm{IG}} - \mu_{\mathcal{R}_{\mathrm{IG}}}}{\sigma_{\mathcal{R}_{\mathrm{IG}}}}, \qquad A_T^{\mathrm{F1}} = \frac{r_T^{\mathrm{F1}} - \mu_{\mathcal{R}_{\mathrm{F1}}}}{\sigma_{\mathcal{R}_{\mathrm{F1}}}} \tag{6}$$

We apply a wide safety cap of ± 50.0 to IG values to prevent numerical instability while preserving the natural variance that GRPO normalization requires. We discuss the sensitivity to this choice in Section 7.

4.4 Curriculum Schedule

We employ a linear curriculum that interpolates the IG reward weight from $\lambda_{\mathrm{IG}}^{\mathrm{init}} = 0.1$ to $\lambda_{\mathrm{IG}}^{\mathrm{final}} = 0.3$ over $S = 200$ training steps:

$$\lambda_{\mathrm{IG}}(s) = \lambda_{\mathrm{IG}}^{\mathrm{init}} + \frac{s}{S}\left(\lambda_{\mathrm{IG}}^{\mathrm{final}} - \lambda_{\mathrm{IG}}^{\mathrm{init}}\right) \tag{7}$$

The combined per-token reward is $r_t = \lambda_{\mathrm{IG}}(s) \cdot r_t^{\mathrm{IG}}$ for intermediate turns and $r_T = r_T^{\mathrm{F1}}$ for the final turn. The curriculum starts with a low IG weight to let the model first learn format compliance, then gradually increases the influence of evidence-reading quality. The F1 weight is fixed at 1.0 throughout training.

5 Experimental Setup

Model and Infrastructure. All experiments use the Qwen2.5-Instruct family [20]. All experiments use Qwen2.5-3B-Instruct (3.09B parameters). Training runs on 2× NVIDIA GPUs (24GB each) with tensor model parallelism of size 2. We use FSDP for distributed training and vLLM [21] for efficient rollout generation. Each training run completes 200 steps (approximately 2.5 hours on our hardware).

Data. We use HotpotQA [5] for multi-hop question answering. The training set contains 1,000 examples; the test set contains 1,000 examples for final evaluation. Evidence documents are pre-indexed locally.

Training Configuration. Key hyperparameters are listed in Table 2. The rollout uses a temperature of 0.6 and a maximum of 3 turns per trajectory. GRPO group size $N = 2$, PPO mini-batch size of 2, and KL coefficient of 0.10. Due to hardware constraints (2×24GB GPUs), group size and batch size are limited to 2; the effect of larger group sizes remains untested. Format violations (missing or broken XML tags, or answers without evidence retrieval) receive a penalty of -2.0.

Evaluation Metrics. We evaluate checkpoints every 50 steps on the full 1,000-example test set. We report two primary metrics:

- **Standard F1**: Token-level F1 score with format-violating outputs scored as 0. This metric is always in $[0, 1]$.
- **Format-penalized reward**: Training reward including the -2.0 penalty for format violations. This can be negative and is reported separately from F1 to avoid confusion.

We also report exact match (EM), with format violations scored as 0 (standard EM in $[0, 1]$), separately from the format-penalized EM reward.[1]

[1] In the original evaluation, format violations received EM $= -2.0$. We rename this column to "Format-penalized EM reward" and report standard EM separately.

Table 2: Training hyperparameters.

Hyperparameter	Value
Model	Qwen2.5-3B-Instruct (3.09B)
Max turns	3
Rollout temperature	0.6
GRPO group size N	2
PPO mini-batch size	2
Learning rate	1×10^{-6}
KL coefficient β	0.10
Entropy coefficient	0.01
IG curriculum range	$0.1 \rightarrow 0.3$
IG safety clip	± 50.0
Normalization mode	separate
Format penalty	-2.0
Max sequence length	3072 tokens
Training steps	200
GRPO Baseline	IG weight = 0 (disabled)

Table 3: Best-checkpoint and final-checkpoint comparison. Standard F1 excludes format penalty (violations scored as 0). Format-penalized reward includes the -2.0 penalty. Format violation rate is computed from the evaluation logs.

Model	Standard F1 (no penalty)	Penalized Reward	FmtViol Rate %	Check-point
Base Qwen2.5-3B	0.252	-0.930	59.1	–
GRPO (best)	0.430	0.008	21.1	Step 50
GRPO (final)	0.000	-2.000	100.0	Step 200
CIGPO (final)	**0.518**	0.272	12.3	Step 200

6 Results

6.1 Best-Checkpoint vs Final-Checkpoint Comparison

Table 3 provides a fair comparison using both best and final checkpoints.

GRPO improves then collapses. GRPO reaches a standard F1 of 0.430 at step 50, but format violations rise from 21.1% (step 50) to 90.5% (step 100), reaching 100% by step 150. The final checkpoint (step 200) produces exclusively format-violating outputs (standard F1 = 0.000).

CIGPO stabilizes and continues improving. CIGPO's standard F1 increases from 0.429 (step 50) to 0.518 (step 200; +21% relative over step 50, +105% over the base model's 0.252). Format violations decrease from 31.5% to 12.3%. We note that standard EM peaks at step 100 (0.242) and then declines to 0.150 at step 200, suggesting that continued training broadens partial answer recall at the expense of exact-match precision (see Section 7).

6.2 Training Dynamics

Figure 2 shows the standard F1 and format violation rate across checkpoints.

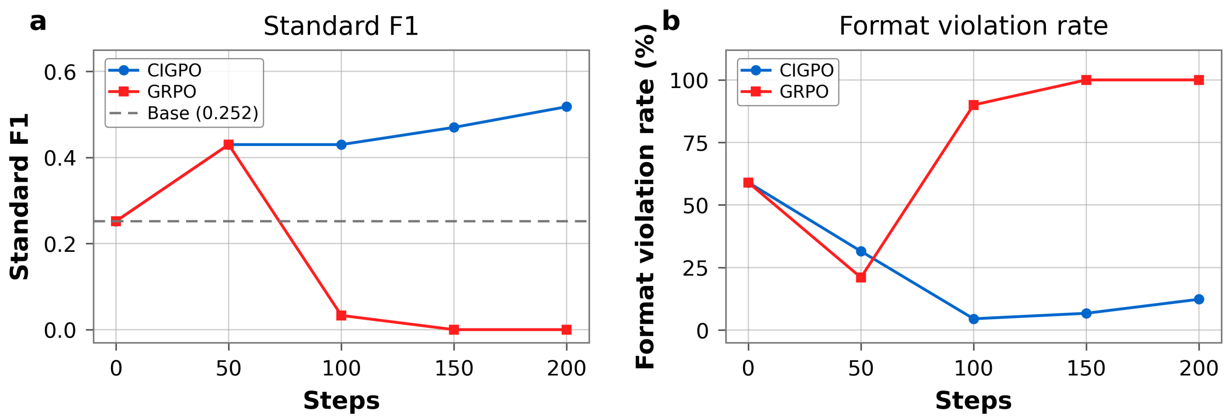

Figure 2: Training dynamics. **Left:** Standard F1 (format violations scored as 0). CIGPO improves from 0.429 to 0.518; GRPO initially reaches 0.430 at step 50 but collapses to 0.000. **Right:** Format violation rate. GRPO violations rise from 21% to 100%; CIGPO violations decrease from 32% to 12%.

Table 4: Full test set results across all checkpoints (HotpotQA, $N = 1000$). Standard F1: format violations $\rightarrow 0$. Penalized reward: violations $\rightarrow -2.0$. Standard EM: violations $\rightarrow 0$.

Checkpoint	Standard F1	Penalized Reward	Standard EM
Base Qwen2.5-3B	0.252	−0.930	0.011
CIGPO Step 50	0.429	−0.201	0.000
GRPO Step 50	0.430	0.008	0.000
CIGPO Step 100	0.432	**0.344**	**0.242**
GRPO Step 100	0.033	−1.781	0.000
CIGPO Step 150	0.471	0.339	0.217
GRPO Step 150	0.000	−2.000	0.000
CIGPO Step 200	**0.518**	0.272	0.150
GRPO Step 200	0.000	−2.000	0.000

6.3 Full Benchmark Results

Table 4 presents the complete per-checkpoint results.

6.4 Collapse Diagnosis

Table 5 provides a detailed breakdown of format violations across checkpoints, computed from the evaluation prediction logs.

At GRPO steps 150–200, all outputs have unclosed XML tags and missing <answer> tags, with zero evidence retrieval—the model produces unstructured token sequences. In contrast, CIGPO maintains high retrieval rates (99–100%) and valid format rates (88–96%) throughout training, with correct answers ($F1 > 0.5$) increasing from 42.1% to 51.3%.

To further characterize the GRPO collapse from the optimizer's perspective, we examine the training log metrics at step 200. Table 6 provides the key metrics that distinguish collapsed GRPO runs from stable CIGPO training, based on metrics directly recorded during training.

The training logs (Table 6) confirm that the GRPO failure corresponds to a zero-advantage lock-in. At step 200, all sampled trajectories receive the minimum format penalty, with critic/score/mean =

Table 5: Collapse diagnosis across checkpoints. Values computed from 1,000-sample evaluation JSONL files. "AnsWoRetr" = answer without evidence retrieval. "HasRetr" = trajectory contains at least one evidence retrieval.

Method	Step	Valid%	FmtViol%	NoAns%	Unclosed%	HasRetr%	HasAns%	F1>0.5%
Base	0	40.9	59.1	58.7	25.9	92.0	41.3	25.2
GRPO	50	78.9	21.1	19.0	0.9	97.4	81.0	42.2
GRPO	100	9.5	90.5	88.2	60.7	96.2	11.8	3.1
GRPO	150	0.0	100.0	100.0	100.0	0.0	0.0	0.0
GRPO	200	0.0	100.0	100.0	100.0	0.0	0.0	0.0
CIGPO	50	68.5	31.5	31.5	0.4	100.0	68.5	42.1
CIGPO	100	95.6	4.4	2.8	0.5	98.5	97.2	41.8
CIGPO	150	93.4	6.6	4.8	0.9	99.1	95.2	45.9
CIGPO	200	87.7	12.3	12.1	0.8	99.6	87.9	**51.3**

Table 6: Training-log diagnosis of GRPO collapse at step 200. All values are taken directly from the GRPO 3B training log at step 200.

Metric	GRPO Step 200	Interpretation
critic/score/mean	-2.000	All sampled trajectories receive the minimum format penalty
critic/rewards/mean	-2.000	Reward distribution collapses to the minimum score
critic/advantages/mean	0.000	No group-relative advantage remains
critic/returns/mean	0.000	Return signal vanishes
actor/pg_loss	0.000	Policy-gradient update vanishes
actor/entropy_loss	≈ 4809	Output distribution becomes highly unstable / high entropy
training/zero_advantage_group_ratio	1.000	All groups have zero advantage
curriculum/ig_weight	0.000	GRPO has no intermediate IG signal

critic/rewards/mean = -2.000. Since group members receive identical rewards, the group-relative advantage vanishes, as reflected by critic/advantages/mean = 0.000 and zero_advantage_group_ratio = 1.000. The policy-gradient loss also becomes 0.000, indicating that the policy receives no effective learning signal to recover from format collapse.

6.5 Per-Turn Information Gain Analysis

To verify that IG rewards are meaningful, we analyze cumulative IG values across all CIGPO trajectories, grouped by outcome. Table 7 shows the results.

Trajectories that produce correct answers have the highest cumulative IG (4.06), while valid wrong trajectories have the lowest IG (2.10). Interestingly, format-violating trajectories still exhibit non-trivial cumulative IG (2.97), suggesting that evidence acquisition and final-format compliance are partially decoupled. This supports the usefulness of IG for evidence acquisition but also motivates future validity gating to prevent malformed trajectories from contributing noisy IG rewards. We caution that correlation does not imply causation—higher IG may reflect easier questions rather than better evidence-reading strategy.

Table 7: Cumulative IG by outcome group (all CIGPO checkpoints, $N = 3972$ parsed trajectories). Correct = standard F1 > 0.5; Partial = $0 <$ F1 ≤ 0.5; Wrong = F1 = 0 (valid format); FormatViol = format violation.

Outcome	N	Mean Cum. IG	Mean Turns	Mean F1
Correct (F1 > 0.5)	1811	4.06	1.5	0.942
Partial ($0 <$ F1 ≤ 0.5)	390	3.28	1.4	0.368
Wrong (F1 = 0, valid)	1251	2.10	1.3	0.000
Format violation	520	2.97	2.0	0.000

7 Discussion

7.1 Why Does GRPO Collapse in Multi-Turn Settings?

The GRPO collapse observed in our HotpotQA setting is not merely a degradation in answer accuracy, but an optimization deadlock. Once format-violating outputs dominate all samples in a GRPO group, all trajectories receive the same minimum penalty of -2.0. The reward standard deviation collapses, the group-relative advantage becomes zero, and the policy-gradient loss vanishes—a deadlock from which the model cannot recover. This is observed directly in the GRPO step-200 training logs (Table 6): score/reward means are -2.000, the zero-advantage group ratio is 1.000, and actor/pg_loss is 0.000. The entropy loss exceeds 4800, consistent with garbled high-entropy outputs that have lost all task structure. The collapse is self-reinforcing: once format violations dominate a group, the policy has no effective learning signal to recover.

CIGPO mitigates this deadlock by supplying sparse turn-level IG rewards at intermediate evidence-reading turns. Even when final-answer rewards become homogeneous, variation in per-turn IG preserves relative advantage signals and reinforces useful evidence acquisition. A single trajectory with relevant evidence reading produces a positive IG-based advantage for its intermediate turns, maintaining a gradient toward effective evidence use.

At a conceptual level, CIGPO can be understood as a **variance-injection mechanism**. By assigning turn-level IG rewards with separate normalization, it ensures that the reward distribution across group members retains non-zero variance even when final-answer rewards are homogeneous. Whether the injected signal is IG, a learned process reward, or a simpler heuristic may be secondary—the essential requirement is that intermediate turns receive a reward signal with sufficient variation to prevent advantage collapse.

7.2 What Component of CIGPO Matters Most?

A limitation of the current study is that we do not isolate which of CIGPO's three components—the IG reward signal, separate per-metric normalization, or the IG-weight curriculum—is primarily responsible for stabilizing training. Under the variance-injection principle, however, these three components have a unified interpretation: (1) the IG reward is the variance *source*, providing a non-constant signal on intermediate turns; (2) separate normalization is the variance *preserver*, preventing F1's $[0, 1]$ scale from drowning IG's nat-scale variance; and (3) the curriculum is the variance *scheduler*, gradually introducing the signal so the model first learns format compliance. The IG signal itself provides turn-level variation that may prevent the reward distribution from collapsing even when final-answer rewards are homogeneous. Controlled ablations (IG-only without curriculum, joint normalization with curriculum, curriculum-only with a constant auxiliary reward) are needed to disentangle the individual contributions. We leave such ablation studies to future work, noting that preliminary experiments with aggressive IG clipping (± 0.5) and joint normalization collapsed IG variance to near zero, suggesting that normalization strategy is at minimum a necessary condition for stability.

7.3 F1–EM Divergence

An interesting pattern in the CIGPO results (Table 4) is that standard F1 and standard EM move in opposite directions late in training: F1 rises from 0.432 (step 100) to 0.518 (step 200), while EM declines from 0.242 to 0.150 over the same period. This divergence suggests that continued training under the combined IG + F1 reward improves the model's ability to retrieve partially correct tokens (boosting F1) while reducing its tendency to produce exact string matches (lowering EM). One possible explanation is that the IG reward guides the model toward reading relevant evidence, which increases token overlap with the ground truth, but the F1-based final reward does not strongly penalize paraphrased or approximate answers—so the model learns to generate correct content without reproducing the exact ground-truth span. This trade-off between recall (F1) and precision (EM) warrants further investigation in future work.

7.4 Limitations

We acknowledge several limitations of the current study:

- **Single dataset.** Only HotpotQA is used. Generalization to other multi-hop QA datasets (2WikiMultihop, MuSiQue), retrieval-augmented generation, or web search agent tasks remains future work.
- **No statistical testing.** Results are reported as point estimates from a single training run per configuration. Confidence intervals, error bars, and significance tests are not reported due to the computational cost of multiple independent training runs; the reported patterns should be interpreted as suggestive rather than confirmatory.
- **Scale sensitivity.** Although CIGPO stabilizes training in our main Qwen2.5-3B HotpotQA setting, preliminary 1.5B runs were unstable for both GRPO and CIGPO. This suggests that smaller models may lack sufficient capacity to reliably follow the evidence-reading protocol and benefit from contextual IG rewards. We therefore restrict our main claims to the 3B setting and leave systematic scaling studies to future work. Larger models (7B–70B) may also exhibit different stability characteristics and may be less prone to collapse even without IG.
- **Small group size.** GRPO group size is 2 due to GPU memory constraints. With $N = 2$, group-relative advantages are binary (± 1 when rewards differ, zero when they match), making GRPO inherently more brittle than with larger groups where advantage magnitudes can reflect reward magnitudes. The zero-advantage deadlock observed here may be partially an artifact of this small group size: any pair of identical rewards (e.g., two format violations) produces zero advantage, whereas with $N \geq 4$, a minority of valid trajectories can still generate non-zero advantages. Larger group sizes could affect both the collapse dynamics and the effectiveness of separate normalization.
- **IG clipping and normalization sensitivity.** In preliminary experiments, aggressive IG clipping (± 0.5) with joint IG–F1 normalization collapsed IG variance to near zero ($\sigma \approx 0$), producing only 2% correct outputs. A wide safety clip (± 50.0) with separate per-metric normalization was necessary for stable training; isolating the individual contribution of clipping threshold and normalization mode remains future work.
- **IG requires ground-truth answers.** IG computation depends on y^* and is only available during training. At inference, the model relies on the learned policy without IG rewards.
- **Reference-model dependence.** Reference-based IG depends on model calibration and may be noisy

for questions with long or ambiguous answers. The quality of the IG signal is bounded by the reference model's ability to assign meaningful probabilities to the ground-truth answer.

- **No validity gating.** The current implementation does not gate invalid or malformed evidence-reading turns from IG normalization. The non-trivial cumulative IG observed for format-violating trajectories in Table 7 suggests that future validity gating may improve the signal-to-noise ratio by excluding malformed evidence-reading turns from IG normalization.

7.5 Future Work

Scaling CIGPO to larger models (7B–70B) and more complex agent tasks would test the generality of variance-injection-based credit assignment. Systematic scaling studies across model sizes (1.5B–70B) are needed to map out the relationship between model capacity and variance-injection effectiveness. Incorporating validity gating to filter malformed turns from normalization could improve the signal-to-noise ratio (see Table 7). Evidence-level hit-rate analysis against HotpotQA supporting facts and investigation of the F1–EM trade-off observed in late-stage training are left for future work. Future work could also isolate the effects of clipping threshold and normalization mode through controlled single-variable experiments. Integrating CIGPO with process reward models [17] could combine learned per-step verification with the principled IG signal.

8 Conclusion

We identified a zero-advantage variance collapse in GRPO-based multi-turn evidence-reading training—all trajectories receive the same -2.0 penalty, advantages vanish, and pg_loss reaches zero—and proposed a variance-injection strategy to prevent it. CIGPO implements this strategy via sparse turn-level contextual information-gain rewards, reaching a standard F1 of 0.518 (from 0.252 base; +0.266). Correct trajectories exhibit substantially higher cumulative IG (4.06 vs. 2.10 for wrong), consistent with IG reflecting successful evidence use. Our claims are restricted to the Qwen2.5-3B HotpotQA setting. Preliminary 1.5B experiments suggest that both GRPO and CIGPO can become unstable at smaller model scales, highlighting model capacity as an important factor for future work. Key open questions—disentangling IG, normalization, and curriculum contributions; testing at larger group sizes; and evaluating on diverse datasets—are left for future work. As a focused empirical study, we hope CIGPO encourages further work on principled credit assignment for multi-turn LLM agent training.

A Qualitative Examples

We present three representative examples from the evaluation prediction logs.

A.1 GRPO Collapsed Example (Step 150)

- **Question**: Billy Preston is an American college basketball player for a team that competes in what?
- **F1 (no fmt penalty)**: 0.0
- **Format-penalized reward**: −2.0
- **Analysis**: The GRPO model at step 150 produces garbled output with no <answer> tag, no coherent evidence retrieval, and unclosed XML tags. The output consists of random token sequences in multiple languages/scripts, representing complete format collapse. This qualitative collapse matches the training-log diagnosis (Table 6): at this stage, all sampled trajectories receive the −2.0 format penalty, the zero-advantage group ratio reaches 1.0, and the policy-gradient loss is 0.0. The base model and CIGPO both answer this question correctly (answer: "NCAA" / "Kansas Jayhawks" / "NCAA Division I").

Trajectory (first 300 chars):

```
<think>.capitalizeia small Zheng ...
[...garbled output across multiple scripts...]
```

A.2 CIGPO Success Example (Step 200)

- **Question**: The Cineplex Odeon Theatres are now operated as which American movie theater chain majority-owned by Chinese conglomerate Dalian Wanda Group?
- **Evidence IDs read**: [4]
- **Predicted answer**: AMC Theatres
- **F1 (no fmt penalty)**: 1.0
- **Analysis**: CIGPO correctly identifies the relevant evidence block, reads it, and extracts the correct answer. The trajectory follows the intended format: think → tool_call → tool_response → think → answer.

Trajectory (truncated):

```
<think>From the question, we need to find out what company owns
Cineplex Odeon Theatres... The evidence ID [4] Cineplex Odeon
Corporation could be helpful...</think>
<tool_call>
{"name": "read", "arguments": {"block_id": 4}}
</tool_call>
[...tool_response with evidence text...]
<think>The evidence shows Cineplex Odeon Corporation was acquired
by AMC Theatres, which is majority-owned by Wanda Group.</think>
<answer>AMC Theatres</answer>
```

A.3 CIGPO Failure Example (Step 200, Valid Format)

- **Question**: What government position was held by the woman who portrayed Shirley in the movie "The Story of Us"?

- **Evidence IDs read**: [1, 1] (re-read same evidence)
- **Predicted answer**: Actress
- **F1 (no fmt penalty)**: 0.0
- **Analysis**: CIGPO maintains valid format and reads evidence, but reads the same evidence block twice (wasting a turn) and produces an incorrect answer. The model correctly identifies the actress but fails to find her government position. This represents a typical CIGPO failure mode: valid format and reasonable evidence selection, but incomplete or incorrect multi-hop reasoning.

Trajectory (truncated):

```
<think>I need to find who portrayed Shirley in "The Story of Us"
and what government position she held. Let me check evidence [1].</think>
<tool_call>
{"name": "read", "arguments": {"block_id": 1}}
</tool_call>
[...tool_response with actress info...]
<think>The evidence shows the actress. Let me re-read for
more details.</think>
<tool_call>
{"name": "read", "arguments": {"block_id": 1}}
</tool_call>
[...same tool_response, noting repeat...]
<think>The evidence only mentions her as an actress, not
government position.</think>
<answer>Actress</answer>
```

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